A POSTNIKOV PRESENTATION OF $\|S^2\|_3$, AND $\pi_3(S^2) = \mathbb{Z}$

Purpose. Informal proof for §2 of `doc/pi3-s2-postnikov-handoff-8.txt` (§3d of the plan). Let $T = \|S^2\|_3$ be the constructed 3-truncation (`SphereTruncation.SphereTrunc.trunc3`). We prove

$$\| \textstyle\sum_{B:\mathrm{Ab}} \sum_{k:K(\mathbb{Z},2)\to_* K(B,4)} \mathrm{Fib}k \simeq_* T\|$$

(Theorem 16); with `MapFib.Final.rep-pteq` it gives the mapping theorem at $X = T$ (Corollary 17), and a two-instance form of `SphereUniversal.Assemble` then gives $\pi_3(S^2) \cong \mathbb{Z}$ from the sphere axiom alone (Theorem 19). Written before the formal proof; numbered statements are meant to be declarations. Foundations as in the development (univalence, images; function extensionality derived); no new axiom. The group B is *not* assumed to be $\mathbb{Z}$: it is Ω^3 of the fibre over the base point, and $B \cong \mathbb{Z}$ is a consequence of the final comparison.

The idea. (1) The constructed 2-truncation $V = \|T\|_2$ is pointed equivalent to $K(\mathbb{Z},2)$: both have the universal property of the 2-truncation of S^2 for maps into 2-types (`Trunc3.up`, `TruncationProps.PtUP.up`, `MapFib.SourceS.Sig.sigma-star`). (2) The fibre F_0 of the unit $u : T \to V$ over the base point is a 2-connected 3-type, hence $K(B,3)$ with $B = \Omega^3 F_0$ (a recognition principle, from the delooping corollaries of `DeloopMaps` and Whitehead's principle). (3) In the library $K(B,4)$ *is* the type of $K(B,3)$-torsors (`EM`: $K(A,n{+}3) = T\,K(A,n{+}2)$, `Deloop`), i.e. of types Y with $\|Y\|$ and a homogeneous structure $\mu : (y : Y) \to K(B,3) \simeq_* (Y,y)$; and the fundamental theorem of `Deloop` identifies $(\mathrm{pt} = t)$ with the underlying type of t. So a map $k : V \to K(B,4)$ with fibres the fibres of u is the same as a homogeneous structure on every fibre of u. (4) The homogeneous structures on the fibres of u form a family of *sets* over the simply connected total space T; so the one structure given at the base point by (2) extends to all of T. This is where simple connectivity removes the twisting: the choices at a point form a set (an $\mathrm{Aut}(B)$-torsor), and a family of sets over a 1-connected type has a section through any given point.

1. Conventions

As in `doc/mapping-fib.pdf`: `isTrunc n X` is "X is $(n-2)$-truncated", `isConn n X` is "X is $(n-2)$-connected"; $K_m^A = K(A,m)$ is `EMMaps.KK[A].K m`, pointed by `pt m`, with $e_m : \Omega K_{m+1}^A \simeq_* K_m^A$ (`KK.e`; e_0 is the band of $K(A,1)$, a group isomorphism); $K_2 = K_2^{\mathbb{Z}}$ (`KZ2`); $F;G$ is `ptcomp F G`, $(F;G)_1 = G_1 \circ F_1$ by definition; $\mathrm{fib}_f(z)$ is `Whitehead.fib`; Fib$k$ is `Fibres.Fib`. (T, t_0, η) is `trunc3`: T is `isTrunc 5` and `isConn 3`, and $\eta^* : \mathrm{Map}_*(T,Y) \to \mathrm{Map}_*(S^2,Y)$ is an equivalence for Y `isTrunc 5` (`up`). $\sigma : S^2 \to_* K_2$ is `MapFib.SourceS.sigma`, and σ^* is an equivalence on maps into every `isTrunc 4` type (`Sig.sigma-star`). $V =$ `(TO three).T` T is `isTrunc 4`, $u =$ `PtUP[four, TO three, T, t0].eta` $: T \to_* V$, $v_0 = u\,t_0$, the fibres of u are `isConn 4` (`TruncOp.conn`), and u^* is an equivalence on maps into `isTrunc 4` types (`PtUP.up`).

2. V is $K(\mathbb{Z},2)$

Definition 1. $r = (\eta^*)^{-1}\sigma : T \to_* K_2$ (`up` at $Y = K_2$, which is `isTrunc 4`, hence 5), so $\eta; r = \sigma$; $\varphi = (u^*)^{-1} r : V \to_* K_2$, so $u;\varphi = r$; $\psi = (\sigma^*)^{-1}(\eta; u) : K_2 \to_* V$, so $\sigma;\psi = \eta;u$.

Lemma 2. *(a)* $\psi;\varphi = \mathrm{id}_{K_2}$*; (b)* $r;\psi = u$*; (c)* $\varphi;\psi = \mathrm{id}_V$*. All equalities of pointed maps.*

Date: 28 October 2026.

Proof. Each by injectivity of an equivalence (`Equivalences.injective`) and associativity (`ptcomp-assoc`). (a) σ^* at $Y = K_2$: $\sigma;(\psi;\varphi) = (\sigma;\psi);\varphi = (\eta;u);\varphi = \eta;(u;\varphi) = \eta;r = \sigma = \sigma;\mathrm{id}$ (`ptcomp-idr`). (b) η^* at $Y = V$ (`isTrunc 4`, hence 5): $\eta;(r;\psi) = (\eta;r);\psi = \sigma;\psi = \eta;u$. (c) u^* at $Y = V$: $u;(\varphi;\psi) = (u;\varphi);\psi = r;\psi = u = u;\mathrm{id}$, by (b). □

Proposition 3. *$\psi_1 : K_2 \to V$ is an equivalence, with inverse φ_1.*

Proof. The first components of Lemma 2(a),(c) are $\varphi_1 \circ \psi_1 = \mathrm{id}$ and $\psi_1 \circ \varphi_1 = \mathrm{id}$ (by definition of `ptcomp`); `happly` and `qinv`. □

Remark 4. We never use the fibres of r: the presentation is built over V, where the fibres of u are 2-connected by construction of the truncation, and transported to K_2 along ψ (§6). (That the fibres of r are 2-connected follows, since $r = \varphi \circ u$ with φ an equivalence, but it is not needed.)

3. Recognising $K(B,3)$

Lemma 5 (delooping an equivalence)**.** *Let (A,a_0), (Y,y_0) be 0-connected, $f : \Omega A \to_* \Omega Y$ with f_1 an equivalence, and suppose* Unloop f *is contractible, with centre $(G, \gamma : f = \Omega G)$. Then G is a pointed equivalence $A \simeq_* Y$.*

Proof. $(\Omega G)_1$ is pointwise equal to f_1 (`happly` of $\mathsf{ap}\,(\lambda w.\, w_1)\,\gamma$), so it is an equivalence (`E.homotopy`, as in `EM1.mu-loop-equiv`); Whitehead's principle (`Whitehead.Pointed.equiv`) concludes. This is the argument of `EM.EM1.mu`, stated once. □

Definition 6 (Ω^3 as an abelian group)**.** For a pointed 3-type (F, f_0) let B_F be the abelian group with carrier $\Omega^3 F = (\mathsf{refl}_{\mathsf{refl} f_0} =_{\Omega^2 F} \mathsf{refl}_{\mathsf{refl} f_0})$, a set since F is `isTrunc 5`; zero refl, addition path composition in $\Omega^2 F$, inverse `sym`; associativity `tassoc`, right unit by definition of `trans`, right inverse `trans-sym-r`, commutativity Eckmann–Hilton (`EH[ΩF].eh`: 2-loops of $(\Omega F, \mathsf{refl})$ commute).

Proposition 7 (recognition)**.** *Let (F, f_0) be `isConn 4` and `isTrunc 5`, $B = B_F$. Then there is a pointed equivalence $e_F : K_3^B \simeq_* (F, f_0)$.*

Proof. Three applications of Lemma 5, climbing $\Omega^2 F \to \Omega F \to F$. Levels: $\Omega^2 F$ is `isConn 2`, `isTrunc 3`; ΩF is `isConn 3`, `isTrunc 4`; F is `isConn 4`, `isTrunc 5` (`conn-id`, `conn-down`; identity types of a truncated type). *Step 1* ($G_1 : K_1^B \simeq_* \Omega^2 F$). $f_1 = (e_0)_1 : \Omega K_1^B \to_* B = \Omega(\Omega^2 F, \mathsf{refl})$, the band, pointed by `band-unit`; it is multiplicative (`band-hom`: addition of B is composition in $\Omega^2 F$ by Definition 6), so Unloop f_1 is contractible by Corollary 12 (`Mult.cor12`: K_1^B 0-connected, $\Omega^2 F$ 1-truncated). *Step 2* ($G_2 : K_2^B \simeq_* \Omega F$). $f_2 = (e_1)_1; G_1 : \Omega K_2^B \to_* \Omega^2 F = \Omega(\Omega F, \mathsf{refl})$, an equivalence (composite); Unloop f_2 contractible by Corollary 11 with $c = 1$ (K_2^B is `isConn 3`, ΩF is `isTrunc 4` = `isTrunc (2+(2+0))`). *Step 3* ($G_3 : K_3^B \simeq_* F$). $f_3 = (e_2)_1; G_2$; Corollary 11 with $c = 2$ (K_3^B is `isConn 4`, F is `isTrunc 5`, hence `isTrunc 6` = `isTrunc (3+(3+0))`). In each step both sides are 0-connected. □

4. Homogeneity of the fibres of u

Write $E_z = \mathrm{fib}_u(z)$ for $z : V$; $F_0 = E_{v_0}$ pointed by $f_0 = (t_0, \mathsf{refl})$. Each E_z is `isConn 4` (`TruncOp.conn`) and `isTrunc 5` (Σ of T and of identity types of the 2-type V). Let $B = B_{F_0}$ (Definition 6), $K_3 = K_3^B$, k_3 its point, and $e_0 = e_{F_0} : K_3 \simeq_* (F_0, f_0)$ (Proposition 7).

Lemma 8 (sections of a family of sets over a 1-connected type)**.** *Let (X, x_0) be `isConn 3`, $Q : X \to U$ a family of sets and $q_0 : Q\,x_0$. Then there is $s : (x : X) \to Q\,x$ with $s\,x_0 = q_0$.*

Proof. Lemma 4 (`lemma4` with $t = 0$, $c = 2$: X is `isConn 3`, each $Q\,x$ is `isTrunc (2+0)`): the pointed sections $\mathrm{PtSec}(X, x_0, Q, q_0)$ form a contractible type; s and $s\,x_0 = q_0$ are the components of its centre. (Suggested by the referee, in place of a unique-choice argument.) □

Lemma 9. *For every $t : T$, $Q\,t = \big(K_3 \simeq_* (E_{ut}, (t, \mathsf{refl}))\big)$ is a set.*

Proof. Corollary 5(ii) (`cor5-equiv` with $t = 1$, $c = 3$): K_3 is `isConn 4` and E_{ut} is `isTrunc (3+2)`. □

Definition 10 (the homogeneous structure)**.** $\mu' : (t : T) \to Q\,t$ is the section of Lemma 8 for $X = T$ (which is `isConn 3`), Q of Lemma 9, and $q_0 = e_0$ (note $E_{ut_0} = F_0$ and $(t_0, \mathsf{refl}) = f_0$ by definition). For $z : V$ and $y = (t, p) : E_z$, $\mu_z(y) : K_3 \simeq_* (E_z, y)$ is $\mu'(t)$ moved along p (J on $p : u\,t = z$). So $\mu_z : \mathrm{Mu}(E_z)$ in the sense of `Deloop.Torsors[`K_3`,`k_3`]`.

5. The classifying map

Let $\mathcal{T}$ = `Deloop.Torsors[`K_3`,`k_3`]`: $\mathrm{Tor} = \sum_{k:\mathrm{BAut}} \|\mathsf{el}k\| \times \mathrm{Mu}(\mathsf{el}k)$, where BAut is the small image of the point K_3 in U and $\mathsf{el} : \mathrm{BAut} \to U$ its inclusion. By definition of `EM` (`EMn.Kstage`, `Step.next`), K_4^B *is* Tor, its point is the centre pt of `Step[1].D`, and the fundamental theorem `D.F` gives equivalences $\tau_t : (\mathrm{pt} = t) \simeq \mathsf{el}(t_1)$ (`F.tr`, `F.equiv`). These by-definition facts are stated once, by `refl`.

Lemma 11 (torsor from a homogeneous structure)**.** *Let $Y : U$ with $\|Y\|$ and $\mu : \mathrm{Mu}(Y)$. There is a torsor $\langle Y, \mu\rangle : \mathrm{Tor}$ and a path $\pi_Y : \mathsf{el}(\langle Y, \mu\rangle_1) = Y$.*

Proof. $\mathcal{T}$`.Fib`$\,Y = \sum_{k:\mathrm{BAut}}(\mathsf{el}k = Y)$ is a proposition (`fib-prop`), inhabited at every point y of Y (`fib-contr Y (`$|y|$`, `μ`, `y`)`), hence inhabited, by induction on $\|Y\|$ (at the universe level of $\mathcal{T}$`.Fib`$\,Y$). Let (k, π_Y) be this element; $\langle Y, \mu\rangle = (k,\ \pi_Y^{-1}{}_*(\|Y\|\text{-witness}, \mu))$, as in `homog`. □

Definition 12 (k')**.** $k'_1 : V \to K_4^B$, $k'_1(z) = \langle E_z, \mu_z\rangle$ (with $\|E_z\|$ from `conn-inh`). Let $c_z = \mathrm{idtoeqv}(\pi_{E_z}) : \mathsf{el}(k'_1(z)_1) \simeq E_z$. The pointing is $k'_0 = \big(\tau^{-1}_{k'_1 v_0}(c^{-1}_{v_0} f_0)\big)^{-1} : k'_1 v_0 = \mathrm{pt}$.

Proposition 13. $\mathrm{Fib}k' = \sum_{z:V}(k'_1 z = \mathrm{pt}) \simeq T$.

Proof. A composite of four equivalences: $\sum_z(k'_1 z = \mathrm{pt}) \simeq \sum_z(\mathrm{pt} = k'_1 z)$ (`sym`, fibrewise), $\simeq \sum_z \mathsf{el}(k'_1(z)_1)$ (τ, fibrewise), $\simeq \sum_z E_z$ (c_z, fibrewise; `Fibres.sigma-equiv` for the three), $\simeq T$ ($(z, (t, p)) \mapsto t$, inverse $t \mapsto (u\,t, (t, \mathsf{refl}))$; J on p). □

The pointing of k' plays no role in Proposition 13: the underlying type of $\mathrm{Fib}k'$ does not depend on it.

6. The presentation

Definition 14. $k_T = \psi; k' : K_2 \to_* K_4^B$, so $(k_T)_1 = k'_1 \circ \psi_1$ by definition.

Proposition 15. $\mathrm{Fib}k_T \simeq T$.

Proof. $\mathrm{Fib}k_T = \sum_{x:K_2}(k'_1(\psi_1 x) = \mathrm{pt}) \simeq \sum_{z:V}(k'_1 z = \mathrm{pt})$ by Σ over the equivalence ψ_1 of bases (`SphereAxiom.sigma-base`, Proposition 3); then Proposition 13. □

Theorem 16 (a Postnikov presentation of T)**.** *For $B = B_{F_0}$, k_T and the equivalence $e : \mathrm{Fib}k_T \simeq T$ of Proposition 15: merely e(base point of $\mathrm{Fib}k_T$) $= t_0$, and every such path makes e a pointed equivalence $\mathrm{Fib}k_T \simeq_* (T, t_0)$. In particular $\|\sum_{B:\mathrm{Ab}} \sum_{k:K_2 \to_* K_4^B} \mathrm{Fib}k \simeq_* (T, t_0)\|$.*

Proof. T is 0-connected (`conn-path`). The truncated Σ over Ab lives in the next universe (Ab has a field in U); the formal statement is the first form, which avoids it (referee), and the mere path is eliminated where it is used, into a proposition (Corollary 17). □

Everything but the last pointing is a construction; nothing is chosen.

7. The mapping theorem at T and $\pi_3(S^2)$

Corollary 17 (`MapThm` at $X = T$)**.** *For $P = \mathrm{Fib}\kappa$ (κ the cup square, `H4Proof.h4`) and $s =$ `MapFib.Final.s`, $s^* : [P,T]_* \to [S^2,T]_*$ is a bijection.*

Proof. Being an equivalence is a proposition (`EquivProp.isEquiv-prop`); eliminate the truncation of Theorem 16 and apply `rep-pteq` with $(A,B,k) = (\mathbb{Z}, B, k_T)$. □

Proposition 18 (Theorem B, two instances)**.** `SphereUniversal.Assemble` *only uses* `Rep.rep` *at $X = P$ (s^* on $[P,P]_*$) and $X = T$ (s^* on $[P,T]_*$). Restated with these two bijections as hypotheses (a module* `Two` *taking T-data, (P,p_0),* `isTrunc 5` *P, s,* `sP`*,* `sT`*), the proof is the same text;* `Assemble` *becomes its instance at* `sP` *=* `rep` *P,* `sT` *=* `rep` *T.*

Theorem 19 ($\pi_3(S^2) = \mathbb{Z}$)**.** *For $(S^2, \mathsf{base}, \mathsf{surf})$ with the sphere axiom, $\pi_3(S^2) \cong \mathbb{Z}$ as groups.*

Proof. Proposition 18 with T from `trunc3`, $P = \mathrm{Fib}\kappa$, s as above, `sP` = `rep-fib` for the triple $(\mathbb{Z}, \mathbb{Z}, \kappa)$, `sT` from Corollary 17, gives $\pi_3 S^2 \cong \pi_3 P$; and $\pi_3 P \cong \mathbb{Z}$ (`PostnikovRealization.Triple.pi3-iso`). □

Remark 20 (on B). The proof gives a group B with merely $T \simeq_* \mathrm{Fib}k_T$, and the comparison then shows $\pi_3 T \cong \pi_3 P \cong \mathbb{Z}$; since $\pi_3 \mathrm{Fib}k_T \cong B$ (`pi3-iso`), *merely* $B \cong \mathbb{Z}$ follows, a posteriori (the pointing of e could be made explicit, which would remove "merely"). It is not used.

8. Correspondence with the formal proof

Status: checked (`./segc sag/pi3-s2-postnikov all`: 71 modules, 0 failed; a fresh process from the directory: 3135 declarations, 0 failed, 1.3 s; every new declaration under 0.02 s). Module `Presentation` (entries in `present.opens`, 103, every line "nothing else opened"; checked against this note by kind, not entry by entry, see its header): `deloop-equiv` (Lemma 5); `O3[F,f0,tF]`: `X1`, `x1`, `A3`, `add`, `set`, `ab` (Definition 6); `Rec[F,f0,cF,tF]`: `cY2`, `cX1`, `tX1`, `f1`, `cell1`, `G1`, `f2`, `cell2`, `G2`, `f3`, `cell3`, `e3` (Proposition 7; step 1 is stated on `EM1.K1`, the definition of $K(B,1)$); `Pres[S2,base,surf,ump,TR]` for any `Trunc3` `TR`: `V`, `v0`, `tV`, `tK2`, `r`, `r-eq`, `phi`, `phi-eq`, `psi`, `psi-eq` (Definition 1), `psi-phi`, `r-psi`, `phi-psi` (Lemma 2), `psi-equiv` (Proposition 3); `E`, `e-conn`, `e-trunc`, `f0`, `RF`, `Q`, `q-set` (Lemma 9), `ext` (Lemma 8), `mu'`, `mu` (Definition 10); `DS` (the instance `EM.Step[1, EMn[B].Kstage 1]`, as the referee suggested), `K4-def`, `pt4-def`, `K3-def` (by `refl`), `tau`; `tor-fib`, `tor`, `tor-el` (Lemma 11); `k1`, `c`, `kp` (Definition 12); `sym-equiv`, `tot-ret`, `tot-fib`, `tot` (Proposition 13); `kT`, `kT-def`; `fib-kT` (Proposition 15); `pres-pt`, `pres` (Theorem 16); `Final[S2,base,surf,ump]`: `sT` (Corollary 17), `A2`, `pi3-S2 : GIso[pi3 S2 base, ab-grp Zab]` (Theorem 19). Module `SphereUniversal`: `Two` (Proposition 18), `Assemble` now its instance (`pi3.opens` updated).

Remaining extras (`present.opens` header): `Rec.cell1` opens the implementation of the band when `band-hom` is checked as the multiplicativity of f_1 (a checker-defect candidate; the text uses only that addition in B is composition); the opaque `TO` is opened to the type of the truncation in the declarations about the fibres E_z (the etaV mechanism of `fibre.opens`); `Final.sT` opens the carrier of classes (extra (ii) of `doc/pi3-s2-postnikov.pdf`).